Paper 6.75 - Causal Code as Next-State Precondition

CQE-paper-06.75
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-06.75.md

Paper 6.75 - Causal Code as Next-State Precondition

Purpose

Paper 6.75 explains how causal code becomes the precondition for Paper 7 and the remaining suite.

Exported State

Paper 6 exports:

```
typed causal edge schema
allowed edge types
allowed statuses
closed proof-support acyclicity rule
open-obligation rule
falsifiers for missing receipt, unknown type, hidden cycle
```

Use in Paper 7

Paper 7 may build a discrete-continuous bridge only if the original discrete receipt remains attached as a causal edge. Interpolation, visualization, or presentation cannot erase the proof dependency that produced the samples.

The allowed transition is:

```
sample receipt -> typed edge -> bridge presentation
```

The disallowed transition is:

```
smooth presentation -> erased discrete receipt
```

Use in Later Papers

Later papers may use causal code for:

- proof graph construction,
- obligation ledgers,
- tool and package receipts,
- product route validation,
- paper-to-paper dependency maps,
- hidden-guess training records.

Every use must preserve edge type, receipt, and status.

Precondition Rule

Before a later paper uses Paper 6, it should be able to answer:

```
What dependency is being represented?
What edge type is used?
What receipt supports the edge?
Is the status open, closed, deferred, or rejected?
Does the closed proof-support subgraph remain acyclic?
```

Conclusion

Paper 6.75 turns causal code into portable proof infrastructure. It lets later papers move, visualize, compose, and deploy results while preserving the dependency record that keeps the suite auditable.